My robotics workbook

Build something, test a robot and explain what you noticed. Four teams of three share two robots. Keep your own ideas and let each teammate try every job.

Robot jobs

Driver: edit and start after the lane is clear. Navigator: predict, place the stopped robot and reset its payload. Observer: watch, measure and record one number, tick or phrase. Keep hands away while the robot moves.

Build jobs

Builder connects pieces. Designer finds parts and checks the plan. Tester tries the model and suggests one improvement. Everyone builds. Switch jobs when your mentor calls time.

Our agreement

We ask before Run, respect the lane, share the parts and controls, and stop when something is wrong. A surprising result gives us a new question. Our models stay in labeled team trays.

Team ID _____ Student IDs A _____ B _____ C _____

Welcome to robotics

Today you will make a robot move and build something for it.

Meet your team

Meet your mentors and teammates. Everyone will get time to build and try the robot. Your mentor will help you with each new step.

Watch a robot in action

Watch the demonstration. You can share something you notice, ask a question or simply watch.

Try an idea together

Choose a signal for your robot and make a removable mascot. Try a small change and see what happens.

Help each other

Take turns with the controls and parts. Check that the lane is clear before starting. Anyone can call stop. Keep hands away while the robot moves.

There is time to explore, ask for help and try again.

Week 1 Meet your robot

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A robot welcome signal

Predict together Number these actions in run order: drive, show Y, wait. What changes if the first and last lines trade places?

Our prediction _____

```
await signal("Y")
await pause(500)
await drive(90)
```

Read this fragment. Use the full project supplied by your mentor.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A removable robot mascot

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about instruction order? What did you notice when the robot ran?

Next idea _____ Saved as _____

Week 2 Delivery to a target

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A 40 cm delivery

Predict together Measure C = ____ cm around our wheel. Predict 180 degrees: ____ cm; 360 degrees: ____ cm. Explain using half and one full turn.

40 cm target: (40 / C) × 360 = _____ wheel degrees. Round at the end.

```
wheel_degrees = 180
await drive(wheel_degrees)
await signal("D")
```

Predict cm = (degrees / 360) × C. Use our measured C and the full supplied project.

Run	Wheel degrees	Predicted cm	Measured cm
1			
2			
3			

Build A cargo tray that keeps its load

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about reaching a target? What did you observe in your distance tests?

Next idea _____ Saved as _____

Week 3 Robot dance studio

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge Push the cone while the robot dances

Predict together How many drives? How many turns? Does the final D repeat? Which indentation tells you?

Our prediction _____

```
for beat in range(3):
    await drive(90)
    await turn(90)
await signal("D")
```

Read this fragment. Use the full project supplied by your mentor.

Run	One change	Prediction	Observed result
1			
2			
3			

Build Attach our cone pusher; test it during moves AND turns

Dance repeats completed ____ / 3 One build change _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about repeating actions? What did you notice during the robot dance?

Next idea _____ Saved as _____

Week 4 Parking assistant

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge Parking beside a wall

Predict together At a 150 mm threshold, classify 220, 150, 90, and -1. Which readings mean near? Which means check the sensor?

Our prediction _____

```
stop_mm = 150
gap = distance_sensor.distance(port.C)
if 0 <= gap <= stop_mm:
    await signal("P")
```

Read this fragment. Use the full project supplied by your mentor.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A garage with a dependable target wall

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about sensors and decisions? What did you observe as the robot neared the wall?

Next idea _____ Saved as _____

Week 5 The floor tells the robot

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A floor marker delivery stop

Predict together If dark reads 12 and light reads 76, is 44 a useful threshold? What happens if the threshold is below both readings?

Our prediction _____

```
threshold = 40
reading = color_sensor.reflection(port.D)
if reading < threshold:
    await signal("M")
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A road with a delivery marker

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about detecting a marker? What did you observe on light and dark surfaces?

Next idea _____ Saved as _____

Week 6 A rescue attachment

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A supply box rescue

Predict together Why might two short side guides keep a cube in front better than one flat beam? How could too much contact slow the robot?

Our prediction _____

```
async def deliver():
    await signal("R")
    result = await seek_marker(threshold)
    print(result)

# inside student_recipe
await deliver()
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A rescue scoop or guide

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn while building your guide? What did you observe when it moved the cube?

Next idea _____ Saved as _____

Week 7 Community robot mission

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A community helper prototype

Predict together Which part of your mission is most likely to fail? What small test could answer that before you run everything?

Our prediction _____

```
mission = "marker"
if mission == "marker":
    result = await seek_marker(threshold)
else:
    result = await park(stop_mm)
print(result)
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A small community mission scene

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn while combining your mission? What did you observe in your first complete test?

Next idea _____ Saved as _____

Week 8 Robot showcase

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A short public demonstration

Predict together Before the demo, predict the stop reason. What will you explain if the robot stops for a limit instead of detecting its target?

Our prediction _____

```
await signal("Y")
result = await seek_marker(threshold)
print("Our result:", result)
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A working exhibit for the showcase

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What can you explain or do now? What did you notice in today's demonstrations?

Next idea _____ Saved as _____

What I can explain now

Student ID _____ Date _____

Answer independently before the Week 8 briefing. Your mentor may read the questions without giving answers.

1. A recipe says: show Y, drive, show D. Which action happens second?

_____

_____

2. wheel_degrees = 180 is changed to wheel_degrees = 360. If everything else stays the same, do you predict more travel, less travel, or exactly the same travel? Why?

_____

_____

3. A loop repeats drive then turn three times. How many drive actions happen?

_____

_____

4. The robot parks when a valid gap is 150 mm or less. Should it park at 100 mm? Should it park at 200 mm?

_____

_____

I can explain a robot program: 1 Not yet / 2 A little / 3 With help / 4 Mostly / 5 Confidently

More robotics? Yes / Maybe / No / Prefer not to answer. My next project idea: _____

Reading the robot recipe

Edit the small student_recipe section in the complete project. Helpers, ports and safety limits are prepared by mentors. Ask before changing them.

Example	Meaning
await signal("Y")	Show a letter
await pause(500)	Wait half a second
await drive(180)	Turn the wheels half a rotation
wheel_degrees = 180	Give a number a name
for beat in range(3):	Repeat the indented actions three times
if reading < threshold:	Choose an action when the comparison is true
await deliver()	Run the named action

Small repairs

Keep quotation marks around letters. Keep the colon after a loop or decision. Preserve indentation. Wheel degrees are not centimeters or the robot’s heading. Measure the real result.